

Macroeconomic Theory and Policy: Lecture W7 (Ch. 18)

What are You Going to Learn (Ch. 18)

- Government spending, taxation, budget deficits, and the debt-GDP ratio nationally and globally
- The government's intertemporal budget constraint
- The economic consequences of budget deficits
- The fiscal problem of the twenty-first century: how to finance rising health expenditures

U.S. Government Spending and Revenue

- In 2018, government spending in the United States was
 - ▶ \$6.7 trillion
 - ▶ 20.0 percent of GDP
 - ▶ more than \$20,500 per person!
- Tax revenues were 16.2 percent of GDP

The U.S. Federal Government Budget, 2018

TABLE 18.1

The U.S. Federal Government Budget, 2018

	Percentage of GDP	Dollars per person
Total expenditures	20.0	12,550
Health (including Medicare)	5.6	3,480
Social Security	4.8	3,020
National defense	3.1	1,930
Income security	2.4	1,510
Net interest	1.6	990
Other	2.6	1,620
Total revenues	16.2	10,170
Individual income taxes	8.2	5,140
Social insurance and retirement receipts	5.7	3,580
Corporate income taxes	1.0	630
Other	1.3	830
Budget deficit	-3.8	-2,380

Source: *Economic Report of the President*, 2018, Table B-19.
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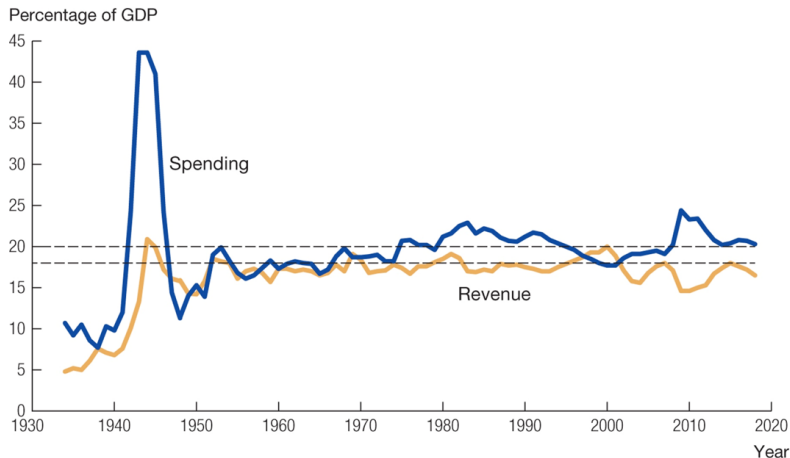
Government Spending and Revenue

- The budget balance: The difference between tax revenues and spending
- A budget surplus: Tax revenues $>$ Spending
- A budget deficit: Tax revenues $<$ Spending
- A balanced budget: Tax revenues $=$ Spending

Government Spending and Revenue

- In 2018, there was a budget deficit equal to 3.8 percent of GDP (0.5 percent in Canada)
 - ▶ The government does not usually sell assets to finance the deficit
 - ▶ Usually, the government borrows from lenders domestically and abroad
- When the government sells bonds, they use the money as a loan
- In 2018, the government issued more than \$750 billion in new bonds (>\$2,300 per person)

U.S. Federal Government Revenue and Spending

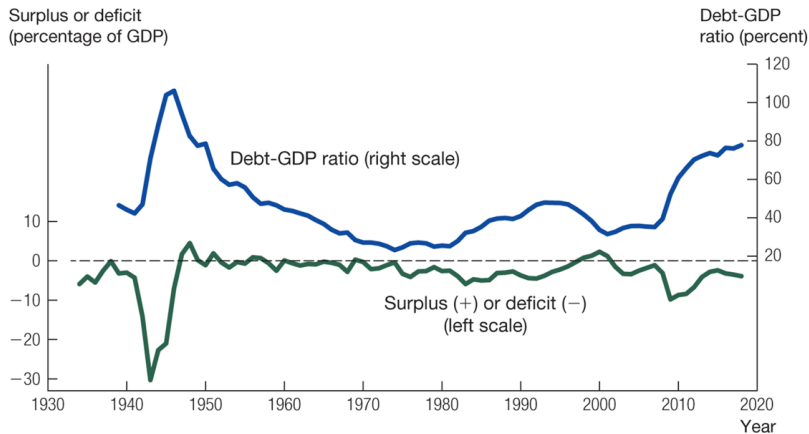


Source: *Economic Report of the President*, 2019, Table B-46.
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The Debt-GDP Ratio

- Government debt: outstanding stock of bonds that have been issued in the past
 - ▶ It is like the capital accumulation function!
- In 2005, the debt-GDP ratio was just over 35 percent
- In 2018, the debt-GDP ratio was nearly 80 percent, a huge increase
- About half of the debt in recent years is owed to foreigners

Federal Debt and Deficits in the United States

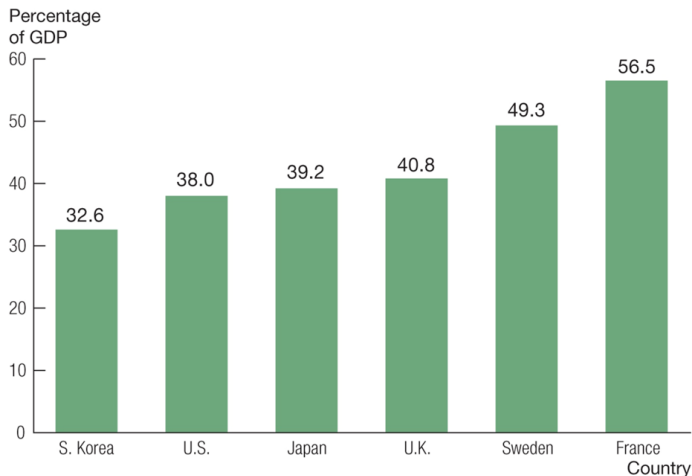


Source: *Economic Report of the President*, 2019, Table B-46.
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International Evidence on Spending and Debt

- Among the richer OECD countries, the United States has a lower than average
 - ▶ government spending-to-GDP ratio
 - ▶ debt-GDP ratio
- Norway is an exception among many rich countries
 - ▶ Negative debt-GDP ratio
 - ▶ Norway benefits from large government-owned petroleum and natural gas reserves, which boost its revenues and lead to budget surpluses
 - ▶ Knowing these revenues will be depleted, the Norwegian government saves these surpluses as assets in a Government Petroleum Fund to finance future government spending

Government Spending around the World, 2017

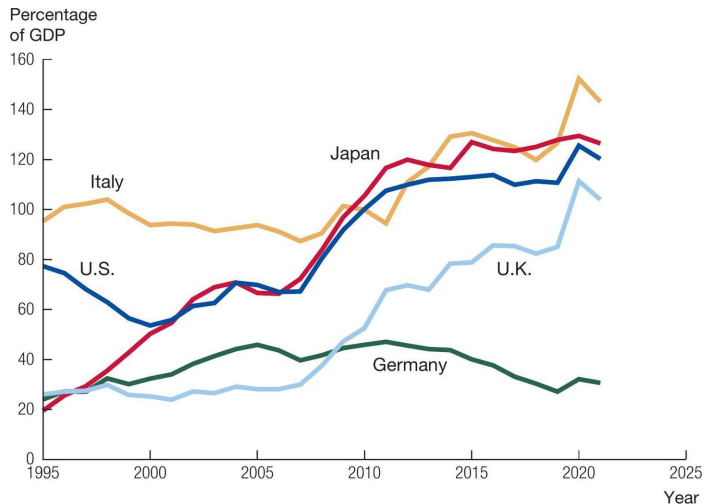


Source: <http://data.oecd.org/gga/>.
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Where would Canada fit?

Debt-GDP Ratios around the World

- In Japan, the ratio has risen sharply since 1995, from a low of about 20 percent to a high of more than 120 percent in 2018



Source: Organisation for Economic Co-operation and Development (OECD) data, data.oecd.org/gga/general-government-financial-wealth.htm.

Debt-GDP Ratios around the World

- The Debt-GDP ratio comprises two components: Debt and GDP
- If the GDP kept growing faster than debt, the debt-GDP ratio could decrease
- When assessing the risk of a country defaulting, it's necessary to project the future growth of the country
 - ▶ Many factor could come into place as we learn in previous chapter

The Government Budget Constraint

- The flow version of the government budget constraint holds in each period
- The sources of funds to the government must equal the uses of funds

$$G_t + Tr_t + (1 + i)B_t = T_t + B_{t+1} + \Delta M_{t+1}$$

where G_t is the gov't purchases, Tr_t is the transfer payment, i is the interest rate, iB_t is the interest payments on debts, T_t is the taxes, ΔB_{t+1} is the New Borrowing and ΔM_{t+1} is the change in money stock

- ΔB_{t+1} and ΔM_{t+1} can be negative

The Government Budget Constraint

- Assume for now, transfer payments are zero ($Tr_t = 0$)

$$B_{t+1} = (1 + i)B_t + G_t - T_t - \Delta M_{t+1}$$

- LHS is the stock of debt at the start of next year
- RHS is the Stock of debt this year, including the required interest payments plus the budget deficit and change in money supply

The Intertemporal Budget Constraint

- Suppose an economy exists for only two periods
 - ▶ Period 1 could represent the country today, and period 2 could represent it in the future
- The budget constraint for period 1 is:

$$B_2 = (1 + i)B_1 + G_1 - T_1 - \Delta M_2$$

- The budget constraint for period 2 is:

$$B_3 = 0 = (1 + i)B_2 + G_2 - T_2$$

- $B_3 = 0$ is because No one is willing to lend as the economy live only for two periods (i.e., period 3 does not exist)
- Assume the money supply won't change between period 2 and 3

The Intertemporal Budget Constraint

- Substitute the budget constraint for period 2 into period 1

$$0 = (1 + i) [(1 + i)B_1 + G_1 - T_1 - \Delta M_2] + G_2 - T_2$$

Divide both side by $(1 + i)$

$$0 = (1 + i)B_1 + G_1 - T_1 - \Delta M_2 + \frac{G_2 - T_2}{1 + i}$$

Rearrange:

$$G_1 + \frac{G_2}{1 + i} = T_1 + \frac{T_2}{1 + i} - (1 + i)B_1 + \Delta M_2$$

- Present discounted value of spending (purchases) and initial debt must equal the present discounted value of tax revenues plus change in money supply

The Intertemporal Budget Constraint

- The government's budget must balance—not period by period, but rather in a PDV sense
- The government must have surpluses in the future (today) to pay off deficits today (future)
- The implication is that the government must increase taxes at some point!
- The logic of the intertemporal budget constraint applies for any number of time periods

How Much Can the Government Borrow?

- The amount the government can borrow is limited by:
 - ▶ The amount it can credibly be expected to pay back
 - ▶ Partly on how large the economy's GDP is
- Recall that Debt-GDP ratio has two components : Debt and GDP
 - ▶ If GDP grows faster than the stock of debt, this is not an issue
- When considering economic consequences of deficits and debts, we must consider
 - ▶ Economic growth
 - ▶ The possibility of high inflation or default
 - ▶ Intergenerational equity

High Inflation and Default

- If the debt-GDP ratio becomes too high:
 - ▶ Lenders worry about the ability of the government to repay
 - ▶ Investors demand higher interest rates
 - ▶ This will itself prevent government borrowing from growing too large – the government have to size down or turn to other sources (e.g., printing money)
- If lenders stop lending (or at low enough interest rate):
 - ▶ The government may print more money to satisfy the budget
 - ▶ This generates inflation

High Inflation and Default

- Default: Government cannot repay certain debts fully (or at less than face value)
- This happened in a number of countries:
 - ▶ Latin America during the early 1980s
 - ▶ Russia in 1998
 - ▶ Argentina in 2002
 - ▶ The euro crisis of recent years, which has already seen Greece “restructure” its debt

Generational Accounting

- When the government borrows, the beneficiaries of borrowing may not be the same people who repay the debt
- Example: World War II
 - ▶ The generation that fought the war made large sacrifices
 - ▶ Future generations benefited from the victory
 - ▶ Future generations paid for the war
- Generational accounting: Calculates the extent to which current policies pass on tax burdens to future generations

Generational Accounting

- A study by Auerbach and Kotlikoff shows
 - ▶ Generations born in the 1980s and early 1990s would pay a lifetime tax rate of about 34 percent of their income
 - ▶ All future generations would have to pay a lifetime tax rate of 71 percent
 - ▶ High and rising debt-GDP ratios imply higher tax rates on future generations
- Achieving equalized lifetime tax rates across generations would require a significant change in current fiscal policy (e.g., tax rate based on age?)
- Another debate is that
 - ▶ Current would benefit from all those borrowings
 - ▶ Future generations are likely to be richer, in part because of the investments in capital, education, and research done by current generation (through borrowing)

The Fiscal Problem of the Twenty-First Century

- In the coming decades, with current policies in place, according to a report from CBO, it is likely that by 2075:
 - ▶ Government spending will rise to 40 percent of GDP
 - ▶ Annual budget deficits could reach 20 percent of GDP
- Our current policies are unsustainable

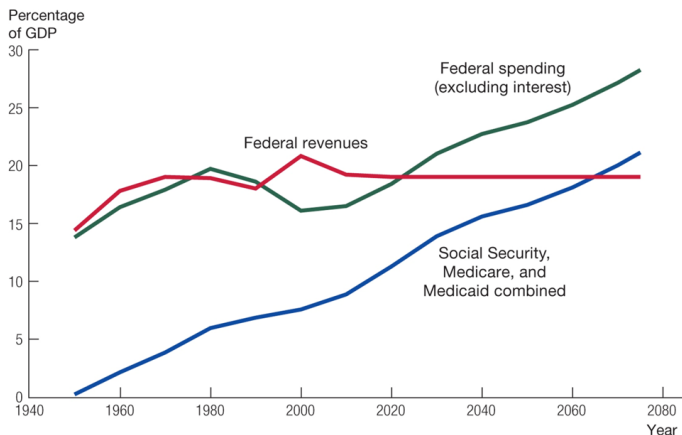
The Problem

It is expected that the expenditure healthcare as a fraction of GDP will increase over time, because:

- Increased generosity of entitlement programs
- More people qualify:
 - ▶ In 1940, there were 8.6 people of working age for every person above 65
 - ▶ In 2000, there were 4.7 people of working age for every person over 65
- Rise in health care expenditures (Medicare/Medicaid)
 - ▶ Health care costs per capita will grow at a rate of 1 percentage point faster than the rate of GDP growth
 - ▶ 7.6 percent of GDP in 2000 to 13.9 percent in 2030 and then to 21.1 percent by 2075

U.S. Federal Spending and Revenues, 1950–2075

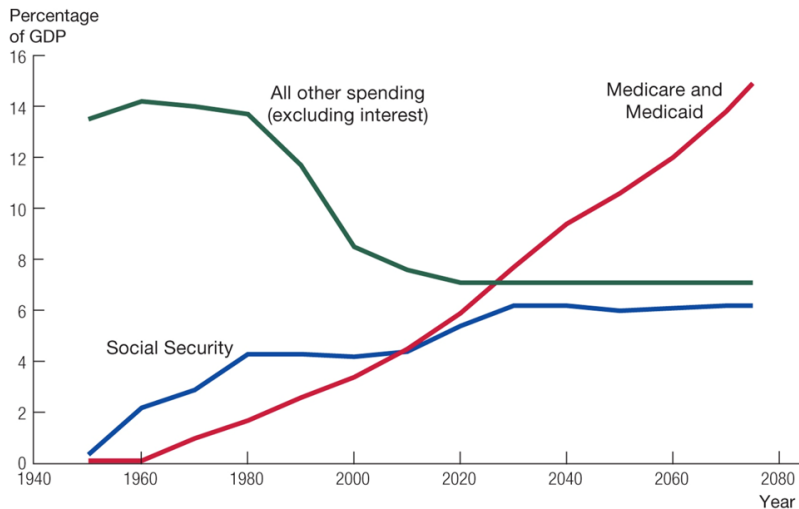
- Health and retirement spending alone will exceed the percentage of GDP collected in taxes (assuming current policies)



Source: Congressional Budget Office, "A 125-Year Picture of the Federal Government's Share of the Economy, 1950 to 2075," Long-Range Fiscal Policy Brief No. 1, 2002.
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Components of Federal Spending, 1950–2075

- The main culprit behind the rise in federal spending is health care then social security



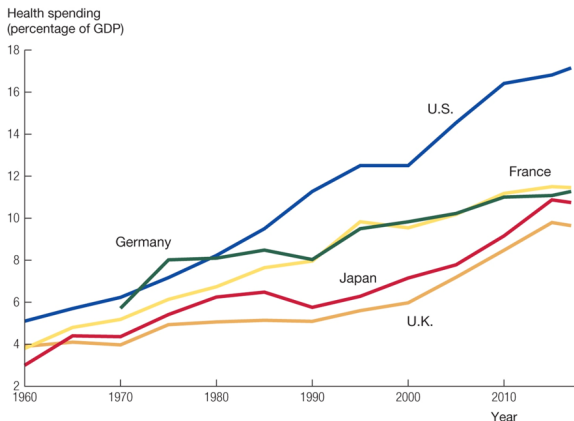
Source: Congressional Budget Office, "A 125-Year Picture of the Federal Government's Share of the Economy, 1950 to 2075," Long-Range Fiscal Policy Brief No. 1, 2002.
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Financing the Social Security Program

- Social security ensures retirees will be able to maintain standards of living
 - ▶ The current tax rate on wage income is 12.4 percent (half paid by employers)
 - ▶ In Canada, the rate is around 11.9 percent
- Financed by an employment tax on wage income
 - ▶ Pay-as-you-go system: Current workers pay the benefits of the current recipients
 - ▶ Fully funded system: Current workers' contribution fully cover the benefits paid out when they retire
- As baby boomers age and retire, this system may not work properly
 - ▶ Workers to retirees ratio is projected to fall from 3.3 to 1 down to a ratio of 2 to 1
 - ▶ Need increased taxes and/or reduced benefits

Possible Solutions?

- Could the budget be balanced? Tax revenues would have to rise by about 9 percent of GDP by 2075
 - ▶ Tax has remain around 20% of GDP for almost entire history of the US
- Health spending is growing in all advanced economies (private and government funded)



Source: OECD Health Data, 2018.
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Possible Solutions?

- Explanations for increasing health care costs:
 - ▶ Expensive medical technologies are raising expenditures
 - ★ New technology is expensive although they are valuable
 - ▶ Waste and fraud in the health system probably does not explain increasing expenses
 - ★ e.g., Medical staff may ask patients to conduct unnecessary tests or procedures
 - ★ Or lack of incentive to actually provide enough services (e.g., roster limit)

Possible Solutions?

- Possible solutions, other than raising taxes, include:
 - ▶ Private health insurance
 - ▶ Mandated savings in individual health-spending accounts
- These may create new problems of their own
 - ▶ Monopoly!?
 - ▶ Scale effect!?

Next Lecture

